**Topic**: Classifying differential equations

**Question**: Which equation is nonlinear?

**Answer choices**:

A $yy' = x^2y - y^2x$

B $y = e^x y' + 2x^3$

C $y' - 2y = e^{x+y}$

D $x^3(y'' - x) = y$

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**Solution**: C

Answer choice A is a linear equation, which we can see if we rewrite it.

$$yy' = x^2y - y^2x$$

$$y' = x^2 - yx$$

$$y' + xy = x^2$$

Answer choice B is a linear equation, which we can see if we rewrite it.

$$y = e^xy' + 2x^3$$

$$e^xy' - y = -2x^3$$

Because the equation $y' - 2y = e^{x+y}$ contains $e^y$, it's non-linear.

Answer choice D is a linear equation, which we can see if we rewrite it.

$$x^3(y'' - x) = y$$

$$x^3y'' - x^4 = y$$

$$x^3y'' - y = x^4$$

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**Topic**: Classifying differential equations

**Question**: Which equation is a third order linear equation?

**Answer choices**:

A $y'' - 2y + 3 = 0$

B $yy''' - 2x = y^2$

C $y''' + 3y = e^x$

D $yy' + ye^x = y^2$

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**Solution**: C

The order of a differential equation is equivalent to the degree of the highest-degree derivative that appears in the equation.

Answer choices B and C contain a third derivative, so they're both third order differential equations.

But the equation $yy''' - 2x = y^2$ in answer choice B is nonlinear,

$$yy''' - 2x = y^2$$

$$yy''' - y^2 = 2x$$

$$y''' - y = \frac{2x}{y}$$

because $Q(x) = 2x/y$. Differential equations are linear when $Q(x)$ is $x$ only, not in $y$.

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**Topic**: Classifying differential equations

**Question**: Which linear equation is homogeneous?

**Answer choices**:

A $y'' + y = y' - 3$

B $y' = e^x y$

C $x^2 y''' + x = y + x^2 y$

D $xy'' - y' = y \sin x + x^2$

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**Solution**: B

A linear equation is homogeneous when $Q(x) = 0$. Rewriting each of the four answer choices in standard form gives

$$y'' - y' + y = -3$$

$$y' - e^x y = 0$$

$$x^2 y''' - (1 + x^2)y = -x$$

$$xy'' - y' - (\sin x)y = x^2$$

Only the equation in answer choice B is homogeneous with $Q(x) = 0$.

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